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Q.1 Non-interacting fermions:

$$V D(\epsilon)\, d\epsilon = g(\epsilon)\, d\epsilon \quad ; \quad \epsilon \le \epsilon_i \le \epsilon + d\epsilon$$

Probability distribution:

$$P_a = \mathcal{Z}^{-1} e^{-\beta(\epsilon_a - \mu)}$$

$$\mathcal{Z} = \prod_a \left(1 + e^{\beta(\mu - \epsilon_a)}\right)$$

(a) $$-\beta\Phi = \ln \mathcal{Z} = \sum_a \ln\left[1 + e^{\beta(\mu-\epsilon_a)}\right]$$

$$\Rightarrow \Phi = -\frac{1}{\beta} \sum_a \ln\left[1 + e^{\beta(\mu-\epsilon_a)}\right]$$

Entropy, $$S = -\frac{\partial \Phi}{\partial T} = k_B \beta^2 \frac{\partial \Phi}{\partial \beta}$$

$$= -k_B \beta^2 \frac{\partial}{\partial \beta}\left[\frac{1}{\beta} \sum_a \ln\left(1 + e^{\beta(\mu-\epsilon_a)}\right)\right]$$

$$= -k_B \beta^2 \left[\left(\frac{-1}{\beta^2}\right) \sum_a \ln\left(1 + e^{\beta(\mu-\epsilon_a)}\right) + \frac{1}{\beta} \sum_a \frac{e^{\beta(\mu-\epsilon_a)} \cdot (\mu - \epsilon_a)}{1 + e^{\beta(\mu-\epsilon_a)}}\right]$$

$$= -k_B \beta^2 \left[\frac{1}{\beta^2} \sum_a \ln\left(\frac{e^{-\beta(\mu-\epsilon_a)}}{1 + e^{-\beta(\mu-\epsilon_a)}}\right) + \frac{1}{\beta} \sum_a (\mu - \epsilon_a)\left(\frac{1}{1 + e^{-\beta(\mu-\epsilon_a)}}\right)\right]$$

$$= -k_B \beta^2 \left[\frac{1}{\beta^2} \sum_a \ln\left(1 - \frac{1}{1 + e^{-\beta(\mu-\epsilon_a)}}\right) + \frac{1}{\beta} \sum_a (\mu - \epsilon_a) \langle n_a \rangle\right]$$

$$= -k_B \beta^2 \left[\frac{1}{\beta^2} \sum_a \ln\left(1 - \langle n_a \rangle\right) + \frac{1}{\beta} \sum_a (\mu - \epsilon_a) \langle n_a \rangle\right]$$

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But, $\langle n_a \rangle = \frac{1}{1+e^{\beta(\epsilon_a-\mu)}} \Rightarrow 1+e^{\beta(\epsilon_a-\mu)} = \frac{1}{\langle n_a \rangle}$

$$\Rightarrow e^{\beta(\epsilon_a-\mu)} = \frac{1}{\langle n_a \rangle} - 1$$

$$\Rightarrow (\epsilon_a-\mu) = \frac{1}{\beta} \ln\left[\frac{1-\langle n_a\rangle}{\langle n_a\rangle}\right] = \frac{1}{\beta}\left[\ln\left[1-\langle n_a\rangle\right] - \ln\langle n_a\rangle\right]$$

$$\therefore S = -K_B\beta^2\left[\frac{1}{\beta^2}\sum_a \ln\left(1-\langle n_a\rangle\right) - \frac{1}{\beta}\sum_a \frac{1}{\beta}\left[\ln\left(1-\langle n_a\rangle\right) - \ln\langle n_a\rangle\right]\langle n_a\rangle\right]$$

$$= -K_B\left[\sum_a\left\{\ln\left(1-\langle n_a\rangle\right) + \langle n_a\rangle \ln\left(1-\langle n_a\rangle\right) + \langle n_a\rangle \ln\langle n_a\rangle\right\}\right]$$

$$S = -K_B\sum_a\left[\left(1-\langle n_a\rangle\right)\ln\left(1-\langle n_a\rangle\right) + \langle n_a\rangle \ln\langle n_a\rangle\right]$$

(b) $$-\beta\Phi = \sum_a \ln\left[1+e^{\beta(\mu-\epsilon_a)}\right] = \int_0^\infty g(\epsilon)\ln\left[1+e^{\beta(\mu-\epsilon)}\right]d\epsilon$$

$$-\beta\Phi = V\int_0^\infty D(\epsilon)\,d\epsilon\,\ln\left(1+e^{\beta(\mu-\epsilon)}\right) \quad -(1)$$

$$\langle N\rangle = \sum_a \frac{1}{1+e^{\beta(\epsilon-\mu)}} = V\int_0^\infty \frac{D(\epsilon)\,d\epsilon}{1+e^{\beta(\epsilon-\mu)}} \quad -(2)$$

(c) $$\beta PV = V\int_0^\infty D(\epsilon)\ln\left[1+e^{\beta(\mu-\epsilon)}\right]d\epsilon$$

$$\Rightarrow P = \frac{1}{\beta}\int_0^\infty D(\epsilon)\ln\left(1+\frac{1}{e^{\beta(\epsilon-\mu)}}\right)d\epsilon$$

$$= \frac{1}{\beta}\int_0^\infty D(\epsilon)\ln\left(\frac{e^{\beta(\epsilon-\mu)}+1}{e^{\beta(\epsilon-\mu)}}\right)d\epsilon$$

$$= -\frac{1}{\beta}\int_0^\infty D(\epsilon)\ln\left[\frac{e^{\beta(\epsilon-\mu)}}{1+e^{\beta(\epsilon-\mu)}}\right]d\epsilon$$

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$$\Rightarrow P = -\frac{1}{\beta}\int_0^\infty D(\epsilon)\ln\left[e^{\beta(\epsilon-\mu)}F(\epsilon)\right]d\epsilon$$

(d) $$\langle E\rangle = \int_0^\infty \epsilon g(\epsilon)F(\epsilon)d\epsilon = V\int_0^\infty F(\epsilon)D(\epsilon)\epsilon\, d\epsilon = \frac{Vm\sqrt{2m}}{\hbar^3\pi^2}\int_0^\infty \epsilon^{3/2}F(\epsilon)d\epsilon$$

$$= \frac{mV\sqrt{2m}}{\hbar^3\pi^2}\int_0^\infty \frac{\epsilon^{3/2}\,d\epsilon}{(e^{\beta\epsilon}e^{-\beta\mu}+1)} = \frac{mV\sqrt{2m}}{\hbar^3\pi^2}\int_0^\infty \frac{\epsilon^{3/2}d\epsilon}{z^{-1}e^{\beta\epsilon}+1}$$

Let, $\beta\epsilon = x \Rightarrow \beta\, d\epsilon = dx$

$$\therefore \langle E\rangle = \frac{mV\sqrt{2m}}{\hbar^3\pi^2}\frac{1}{\beta^{5/2}}\int_0^\infty \frac{x^{3/2}\,dx}{z^{-1}e^{x}+1}$$

$$\langle E\rangle = \frac{mV\sqrt{2m}}{\hbar^3\pi^2}\frac{1}{\beta^{5/2}}\left[\frac{-3}{4}\sqrt{\pi}\,Li_{5/2}(-z)\right], \quad Li_n(x) \text{ is polylogarithm} \quad \text{—— } \circledast$$

But,

$$P = \frac{1}{\beta}\frac{m\sqrt{2m}}{\hbar^3\pi^2}\int_0^\infty \epsilon^{1/2}\ln\left(1+ze^{-\beta\epsilon}\right)d\epsilon$$

$$= \frac{1}{\beta}\frac{m\sqrt{2m}}{\hbar^3\pi^2}\frac{1}{\beta^{3/2}}\int_0^\infty x^{1/2}\ln\left(1+ze^{-\beta\epsilon}\right)d\epsilon$$

$$P = \frac{m\sqrt{2m}}{\hbar^3\pi^2\beta^{5/2}}\left[\frac{-1}{2}\sqrt{\pi}\,Li_{5/2}(-z)\right] \quad \text{—— } \circledast\circledast$$

Comparing $\circledast$ and $\circledast\circledast$;

$$PV = \frac{2}{3}\langle E\rangle$$

$$\Rightarrow \boxed{\langle E\rangle = \frac{3}{2}PV}$$

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(e) $\langle N \rangle = \int_0^\infty g(\epsilon)\, F(\epsilon)\, d\epsilon$

At $T = 0K$; $F(\epsilon) = 1$

$$\therefore \langle N \rangle = \frac{mV}{\hbar^3 \pi^2} \sqrt{2m} \int_0^{\epsilon_F} \epsilon^{1/2}\, d\epsilon$$

$$= \frac{mV\sqrt{2m}}{\hbar^3 \pi^2} \left(\frac{2}{3} \epsilon_F^{3/2} \right) \quad -(1)$$

and,

$$\langle E \rangle = \int_0^\infty g(\epsilon)\, \epsilon\, F(\epsilon)\, d\epsilon$$

At $T = 0K$; $F(\epsilon) = 1$

$$\therefore \langle E \rangle = \frac{mV}{\hbar^3 \pi^2} \sqrt{2m} \int_0^{\epsilon_F} \epsilon^{3/2}\, d\epsilon$$

$$= \frac{mV}{\hbar^3 \pi^2} \sqrt{2m} \left(\frac{2}{5} \epsilon_F^{5/2} \right) \quad -(2)$$

From (1) and (2):

$$\langle E \rangle = \frac{2}{5} \left(\frac{3\langle N \rangle}{2} \right) \epsilon_F$$

$$\boxed{\langle E \rangle = \frac{3}{5} \langle N \rangle \epsilon_F}$$

(f) $$\frac{3}{2} PV = \langle E \rangle = \frac{3}{5} \langle N \rangle \epsilon_F$$

$$\Rightarrow \boxed{P = \frac{2}{5} \frac{\langle N \rangle}{V} \epsilon_F}$$

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8.2 (a) $\langle E \rangle = \int_0^\infty g(\epsilon)\, \epsilon\, F(\epsilon)\, d\epsilon = V \int_0^\infty D(\epsilon)\, \epsilon\, F(\epsilon)\, d\epsilon$

Given,

$$\int_0^\infty d\epsilon\, F(\epsilon)\, \phi(\epsilon) = \int_0^\mu d\epsilon\, \phi(\epsilon) + \frac{\pi^2}{6} (k_B T)^2 \phi'(\mu) + \frac{7\pi^4}{360} (k_B T)^4 \phi'''(\mu)$$

(a) $u = \frac{\langle E \rangle}{V} = \int_0^\infty D(\epsilon)\, \epsilon\, F(\epsilon)\, d\epsilon$

$$u = \int_0^\mu d\epsilon\, \epsilon\, D(\epsilon) + \frac{\pi^2}{6} (k_B T)^2 \left[D(\mu) + \mu D'(\mu) \right]$$

and,

$$n = \frac{N}{V} = \frac{\int_0^\infty g(\epsilon) F(\epsilon) d\epsilon}{V} = \int_0^\infty D(\epsilon)\, F(\epsilon)\, d\epsilon$$

$$\Rightarrow n = \int_0^\mu d\epsilon\, D(\epsilon) + \frac{\pi^2}{6} (k_B T)^2 D'(\mu)$$

(b) $$n = \int_0^{\epsilon_F} d\epsilon\, D(\epsilon) + \int_{\epsilon_F}^\mu d\epsilon\, D(\epsilon) + \frac{\pi^2}{6} (k_B T)^2 D'(\mu)$$

As $T \to 0$, $\int_0^{\epsilon_F} d\epsilon\, D(\epsilon) = n$, and to make n constant as T varies, we must have,

$$\int_{\epsilon_F}^\mu d\epsilon\, D(\epsilon) = -\frac{\pi^2}{6} (k_B T)^2 D'(\mu)$$

$(k_B T)^2 << 1 \Rightarrow \int_{\epsilon_F}^\mu d\epsilon\, D(\epsilon)$ is very small.

$$\therefore (\mu - \epsilon_F) D(\epsilon_F) = -\frac{\pi^2}{6} (k_B T)^2 D'(\mu)$$

To the lowest order $D'(\mu) \simeq D'(\epsilon_F)$

$$\Rightarrow \mu(T) = \epsilon_F - \frac{\pi^2}{6} \frac{(k_B T)^2}{D(\epsilon_F)} D'(\epsilon_F)$$

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$$u = \int_0^{\epsilon_F} d\epsilon\, \epsilon\, D(\epsilon) + \int_{\epsilon_F}^{\mu} d\epsilon\, \epsilon\, D(\epsilon) + \frac{\pi^2}{6}(K_B T)^2 \left[D(\mu) + \mu D'(\mu)\right]$$

$$u = u_0 + (\mu - \epsilon_F)\,\epsilon_F\, D(\epsilon_F) + \frac{\pi^2}{6}(K_B T)^2 \left[D(\mu) + \mu D'(\mu)\right]$$

$\mu - \epsilon_F$ is small, so we can evaluate the third term at $\mu = \epsilon_F$.

$$\therefore u = u_0 + \frac{\pi^2}{6}\epsilon_F (K_B T)^2 D'(\epsilon_F) + \frac{\pi^2}{6}(K_B T)^2 D(\epsilon_F) + \frac{\pi^2}{6}(K_B T)^2 \epsilon_F D'(\epsilon_F)$$

$$\Rightarrow u = u_0 + \frac{\pi^2}{6}(K_B T)^2 D(\epsilon_F)$$

(c) $$C_v = \frac{C_V}{V} = \frac{1}{V}\left(\frac{\partial E}{\partial T}\right)_V = \left(\frac{\partial u}{\partial T}\right)_V$$

$$\Rightarrow C_v = \frac{\pi^2}{6} 2 K_B^2 T\, D(\epsilon_F) = \frac{\pi^2}{3} K_B^2 T\, D(\epsilon_F)$$

$$\Rightarrow C_v = \gamma T^k \;;\; k = 1 \text{ & } \gamma = \frac{\pi^2}{3} K_B^2 D(\epsilon_F)$$

(d) In 3-D, $D(\epsilon) = C\sqrt{\epsilon}$; $C = \frac{m\sqrt{2m}}{\hbar^3 \pi^2}$ is constant.

$$\mu(T) = \epsilon_F - \frac{\pi^2}{6}\frac{(K_B T)^2}{C\sqrt{\epsilon_F}}\left(\frac{1}{2}\frac{C}{\sqrt{\epsilon_F}}\right) = \epsilon_F - \frac{\pi^2}{12}\frac{(K_B T)^2}{\epsilon_F}$$

$$u = u_0 + \frac{\pi^2}{6}(K_B T)^2 C\sqrt{\epsilon_F} = u_0 + \frac{\sqrt{2m^3}\,\pi^2 (K_B T)^2}{6\hbar^3 \pi^2}\epsilon_F^{1/2}$$

and, $$\gamma = \frac{\pi^2}{3} K_B^2 \frac{\sqrt{2m^3}}{\hbar^3 \pi^2}\epsilon_F^{1/2}$$

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